

Quiz 1

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1. Out of KCl and CaO, which one has a lower melting point? Explain how you made each Choice.

The melting point of an ionic compound is directly proportional to its lattice energy—the energy required to break the electrostatic attractions between ions in a solid structure.

KCl (Lower Melting Point): Contains ions with single charges. The lower charge density leads to weaker electrostatic forces, a lower lattice energy, and consequently, a lower melting point.

CaO (Higher Melting Point): Contains ions with double charges. According to Coulomb's Law, higher charges result in much stronger attractive forces and a significantly higher lattice energy, requiring more energy to break the bonds.

2. If an intrinsic semiconductor is doped with a trivalent impurity, how will the conductivity change upon temperature change? Explain with logic.

When an intrinsic semiconductor is doped with a trivalent impurity (e.g., Boron, Aluminum, Gallium), it becomes a p-type semiconductor. The conductivity of this p-type semiconductor increases as temperature increases.

Trivalent Impurity: Trivalent impurities have three valence electrons, while the intrinsic semiconductor (e.g., Silicon) has four.

Hole Generation: When a trivalent atom replaces a Silicon atom, it creates a vacancy in the bond structure—a "hole"—because it lacks one electron to complete the four covalent bonds.

Acceptor Levels: These impurity atoms create allowed energy levels just above the valence band, called "acceptor levels." Even at low temperatures, electrons from the valence band can fill these holes, leaving free holes in the valence band for conduction.

Low to Room Temperature (Extrinsic Range): As temperature increases, more acceptor atoms become ionized, and more holes are produced in the valence band, increasing. While mobility might decrease slightly due

to scattering, the rapid increase in carrier concentration dominates, causing an overall increase in conductivity.

High Temperature (Intrinsic Range): At very high temperatures, thermal energy becomes strong enough to break a large number of intrinsic Silicon bonds, generating a huge number of electron-hole pairs. These thermally generated carriers exceed the number of holes produced by doping. At this stage, the semiconductor behaves like an intrinsic semiconductor, where the concentration of carriers increases exponentially, significantly increasing conductivity